

KATSUURA, Japan

WMO: 476740

Lat: 35.1508N

Lon: 140.3122E

Elev: 14

StdP: 101.16

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.5	1.4	-11.9	1.4	4.1	-10.2	1.6	4.2	11.3	12.2	9.5	10.9	3.4	340	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.7	29.8	26.0	29.1	25.7	28.3	25.3	26.7	28.8	26.2	28.2	25.7	27.7	3.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.0	21.4	28.3	25.6	20.8	27.8	25.1	20.3	27.4	83.7	29.0	81.6	28.4	79.7	27.8	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.2	8.4	7.1	-1.5	31.6	1.2	0.9	-2.4	32.3	-3.1	32.8	-3.7	33.2	-4.6	33.9		
			-4.0	27.3	1.0	0.6	-4.8	27.7	-5.4	28.1	-6.0	28.4	-6.8	28.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	6.8	7.1	10.0	14.0	18.0	20.9	24.2	26.2	23.7	19.1	14.2	9.4
	DBStd	7.02	2.44	2.76	3.12	2.83	2.14	1.79	1.94	1.67	2.51	2.54	2.83	2.83
	HDD10.0	286	104	90	41	4	0	0	0	0	0	2	45	
	HDD18.3	1518	357	315	259	131	31	3	0	0	1	20	127	276
	CDD10.0	2546	6	8	39	125	249	327	440	502	412	283	127	28
	CDD18.3	736	0	0	0	1	21	80	182	243	162	45	2	0
	CDH23.3	4261	0	0	0	0	6	85	972	2092	1003	103	0	0
	CDH26.7	679	0	0	0	0	0	2	114	419	142	2	0	0
Wind	WSAvg	3.3	3.5	3.6	3.7	3.7	3.3	2.9	2.8	2.9	3.1	3.3	3.3	3.5
Precipitation	PrecAvg	1894	97	99	163	152	164	207	155	115	236	254	156	86
	PrecMax	2314	226	214	275	300	296	421	438	320	490	752	435	258
	PrecMin	1292	2	14	22	69	86	80	6	1	76	39	31	6
	PrecStd	297	59	50	56	57	51	83	114	92	109	160	96	56
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.9	17.9	18.7	21.0	24.1	26.5	30.2	31.1	30.3	26.6	22.6	20.3
		MCWB	14.0	15.4	15.1	16.8	19.8	22.0	25.6	26.6	25.8	22.9	18.8	18.0
	2%	DB	15.2	16.1	17.6	19.9	23.0	25.0	28.9	30.2	29.1	25.5	21.5	18.3
		MCWB	11.9	13.5	14.5	16.8	19.1	21.7	25.4	26.4	25.6	22.4	17.9	14.9
	5%	DB	13.4	14.6	16.8	19.2	22.1	24.3	28.0	29.5	28.2	24.6	20.6	16.5
		MCWB	9.2	11.4	13.9	16.2	18.7	21.4	25.0	26.0	25.0	21.4	17.1	12.6
	10%	DB	11.7	12.8	15.8	18.5	21.4	23.6	27.0	28.8	27.4	23.5	19.5	14.6
		MCWB	7.2	8.8	12.7	15.6	18.4	21.1	24.6	25.7	24.6	20.1	15.9	10.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.1	16.0	16.7	19.2	21.3	23.6	26.7	27.5	26.7	24.3	20.3	18.7
		MCDB	16.5	17.4	17.8	19.9	22.7	25.0	28.9	29.9	28.9	25.6	21.7	20.1
	2%	WB	12.4	14.0	15.5	18.0	20.5	22.8	26.0	27.0	26.1	23.3	19.0	15.6
		MCDB	14.7	15.7	17.1	19.3	21.9	23.9	27.9	29.2	28.1	25.0	20.8	17.8
	5%	WB	9.7	11.7	14.4	17.0	19.9	22.2	25.5	26.5	25.6	22.0	17.7	13.1
		MCDB	12.6	14.3	16.4	18.6	21.3	23.5	27.3	28.7	27.6	23.9	20.0	16.0
	10%	WB	7.8	9.3	13.1	16.2	19.2	21.7	25.0	26.1	25.1	20.8	16.4	11.0
		MCDB	11.0	12.1	15.5	18.1	20.8	23.1	26.6	28.2	27.0	22.9	18.9	13.9
Mean Daily Temperature Range	5% DB	MDBR	7.5	7.5	7.2	6.4	5.5	4.6	4.3	4.7	5.2	5.9	6.9	7.3
		MCDBR	9.1	9.2	7.3	6.5	6.2	5.4	5.3	5.4	6.5	7.6	8.9	
		MCWBR	7.8	8.5	6.6	5.0	4.0	3.2	2.8	2.7	3.0	4.6	6.0	7.4
	5% WB	MCDBR	8.8	9.0	7.0	5.2	5.1	4.2	4.8	4.8	5.0	5.8	7.1	8.6
		MCWBR	8.2	8.7	7.2	4.9	4.1	2.9	2.7	2.7	3.2	5.1	6.5	8.1
Clear-Sky Solar Irradiance	taub	0.388	0.440	0.504	0.547	0.558	0.571	0.568	0.546	0.495	0.479	0.432	0.390	
	taud	2.229	2.081	1.930	1.873	1.907	1.948	2.002	2.055	2.172	2.132	2.174	2.234	
	Ebn at Noon	789	784	765	755	753	741	740	748	768	737	731	759	
	Edh at Noon	109	142	179	199	195	187	176	164	139	134	115	103	
All-Sky Solar Radiation	RadAvg	2.59	3.03	3.90	4.65	5.07	4.50	5.16	5.27	4.00	2.97	2.39	2.29	
	RadStd	0.21	0.35	0.29	0.48	0.52	0.44	0.85	0.45	0.34	0.30	0.22	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+0.37	+0.46	+0.54	N/A	N/A	+94	+65
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page